Doyeon (Doy) **Kim**

PH.D. CANDIDATE · LEARNING SCIENCES · EDUCATIONAL TECHNOLOGY

#958 Educational Sciences, 1025 W. Johnson St., Madison, WI 53706

doy.kim@wisc.edu | doy-kim-research | doy-kim

Education

University of Wisconsin-Madison *Madison, WI*

PH.D. IN EDUCATIONAL PSYCHOLOGY – LEARNING SCIENCES *Jan 2023–Present*

- Minor: Quantitative Educational Research
- Advisor: Mitchell J. Nathan

University of Wisconsin-Madison *Madison, WI*

M.SC. IN EDUCATIONAL PSYCHOLOGY – LEARNING SCIENCES *Aug 2019–Dec 2022*

- Advisor: Mitchell J. Nathan

Seoul National University *Seoul, South Korea*

M.SC. IN MATHEMATICS EDUCATION *Mar 2016–Feb 2019*

- Coursework Emphasis: Cognitive Science
- Advisor: Oh Nam Kwon

Seoul National University *Seoul, South Korea*

B.SC. IN MATHEMATICS EDUCATION *Mar 2009–Feb 2016*

- Advisor: Oh Nam Kwon

Research Projects and Professional Experience

Department of Educational Psychology – University of Wisconsin-Madison *Madison, WI*

PH.D. CANDIDATE IN LEARNING SCIENCES *Sep 2024–Present*

- \textit{iGRASP: Immersive Geometric Reasoning through Action and Spatial Practice}: Investigates how body movements affect geometry reasoning and learning within an Extended Reality (XR) environment; addresses a critical gap in the literature regarding body movement and mathematics learning in XR
- \textit{iGRASP: Immersive Geometric Reasoning through Action and Spatial Practice}: Employs an intra-medium randomized 2×2 factorial experiment manipulating sensorimotor engagement and conceptual congruence (High/Low) of physical actions on virtual geometric shapes
- \textit{iGRASP: Immersive Geometric Reasoning through Action and Spatial Practice}: Uses structural equation modeling to estimate treatment effects and model the cognitive processes of mathematics learning in the XR environment
- \textit{GRAID: Geometric Reasoning Analysis through Integrated Datasets}: Examines how physical actions and gestures affect mathematical thinking in non-technology-supported environments
- \textit{GRAID: Geometric Reasoning Analysis through Integrated Datasets}: Utilizes Integrative Data Analysis (IDA) merging raw datasets from nine existing randomized-controlled experiments (N=774); applies Item Response Theory to harmonize measures across datasets
- \textit{GRAID: Geometric Reasoning Analysis through Integrated Datasets}: Applies structural equation modeling (MIMIC framework) to model cognitive processes behind physical actions and geometric reasoning

WestEd *Remote*

DOCTORAL RESEARCH INTERN, CENTER FOR TEACHING AND LEARNING *Jun 2023–Present*

- \textit{New Virtual Reality Technology to Enhance Students' Algebra Knowledge (US DOE S411C230211)}: Evaluated the efficacy of Curio — a VR app designed to enhance engagement and learning in Geometry for high-need and rural students in Alabama
- \textit{New Virtual Reality Technology to Enhance Students' Algebra Knowledge (US DOE S411C230211)}: Roles: instrument development; data collection through interviews, online surveys, and site visits; data analysis; report writing
- \textit{Illustrative Mathematics Evaluation 2023 (Bill & Melinda Gates Foundation INV-051124)}: Evaluated the efficacy of Illustrative Math in improving student math outcomes and explored implementation contexts that lead to better outcomes
- \textit{Illustrative Mathematics Evaluation 2023 (Bill & Melinda Gates Foundation INV-051124)}: Gathered assessment data from state education agencies; visualized data collection progress; transformed raw data for a quasi-experimental study

Wisconsin Center for Education Research – University of Wisconsin-Madison
Madison, WI

GRADUATE RESEARCHER — *Aug 2019–Present*

- \textit{EXCEL: Exploring Collaborative Embodiment for Learning (US DOE R305A200401)}: Explored how innovative action-based technologies (VR, AR, large touchscreen boards) can enhance collaborative mathematical reasoning among high school students
- \textit{EXCEL: Exploring Collaborative Embodiment for Learning (US DOE R305A200401)}: Collected and analyzed data; curated findings for effective communication; presented results at multiple conferences and in a journal article
- \textit{How Dynamic Gestures and Directed Actions Contribute to Mathematical Proof Practices (US DOE R305A160020)}: Investigated the impact of a motion-capture game on high school students' mathematical reasoning
- \textit{How Dynamic Gestures and Directed Actions Contribute to Mathematical Proof Practices (US DOE R305A160020)}: Partnered with local schools to enhance in-classroom studies; pioneered a remote data collection protocol during COVID-19; analyzed data and shared findings at multiple conferences

Department of Educational Psychology – University of Wisconsin-Madison
Madison, WI

TEACHING ASSISTANT — *Aug 2021–May 2022; Aug 2024–Aug 2025*

- \textit{EDPSYCH 301: How People Learn}: Provided personalized guidance and constructive feedback to students through targeted support during assignments and office hours

Social Science Korea
Seoul, South Korea

GRADUATE RESEARCHER — *Nov 2017–Feb 2019*

- \textit{EDPSYCH 301: How People Learn}: Provided personalized guidance and constructive feedback to students through targeted support during assignments and office hours

Center for In-Service Teacher Education
Seoul, South Korea

GENERAL MANAGER — *Jun 2016–Oct 2017*

- Examined how a dynamic digital environment impacts elementary school students' understanding of math concepts
- Independently initiated, managed, and communicated all project aspects — from conceptualization and data collection to writing and presentation

EBS Math – Educational Broadcasting System
Seoul, South Korea

EDUCATIONAL CONTENT RESEARCHER — *Apr 2013–Jun 2014*

- Recruited and developed a cross-functional team of four
- Implemented 50+ professional development programs over 1500+ hours for 7000+ in-service teachers; collected feedback through surveys on all programs
- Overhauled the website and launched a social media account to improve accessibility to program information
- Raised institutional evaluation grade from B to A (2016–2018 Institutional Evaluation)

Publications

PEER-REVIEWED

1. Kim, D., Schenck, K. E., Walkington, C., & Nathan, M. J. (2026). How interaction type with dynamic geometry software shapes deductive geometric reasoning. In K. R. Leatham, B. E. Peterson, D. Teuscher, & D. Siebert (Eds.), *Proceedings of the 48th annual meeting of the north american chapter of the international group for the psychology of mathematics education (PME-NA 48)*.
2. Kim, D., Schenck, K. E., Walkington, C., & Nathan, M. J. (2026). Embodied cognitive pathways from dynamic geometry software access to deductive geometric reasoning. In K. Peppler, M. Ito, K. S. Tekinbas, & M. Dahn (Eds.), *Proceedings of the 20th international conference of the learning sciences (ICLS 2026)*. International Society of the Learning Sciences.
3. Kim, D., Kim, C., & Nathan, M. J. (2026). What makes actions matter in virtual reality? Examining the effects of conceptual congruence and sensorimotor engagement on geometric reasoning. *Proceedings of the 12th Immersive Learning Research Network Conference (iLRN 2026)*.
4. Kim, C., Swart, M. I., Beier, J. P., Xia, F., Grondin, M. M., Kim, D., & Nathan, M. J. (2025). A comparison of teachers and students in categorizing and representing geometric conjectures using AR simulations. *2025 11th International Conference of the Immersive Learning Research Network (iLRN)*.
5. Kim, C., Swart, M. I., Xia, F., Kim, D., & Nathan, M. J. (2025). Exploring teachers' perspectives on the affordances and constraints of augmented reality in geometry education. *2025 11th International Conference of the Immersive Learning Research Network (iLRN)*.
6. Kim, D., Schenck, K. E., Xia, F., Swart, M. I., Walkington, C., & Nathan, M. J. (2025). The effect of dynamic geometry simulation access on embodied geometric reasoning: An experiment. *2025 AERA Annual Meeting*.

7. Schenck, K. E., Kim, D., Xia, F., Swart, M. I., Walkington, C., & Nathan, M. J. (2024). Exploring an interactive technology for supporting embodied geometric reasoning. In R. Lindgren, T. Asino, E. A. Kyza, C.-K. Looi, D. T. Keifert, & E. Suárez (Eds.), *Proceedings of the 18th international conference of the learning sciences (ICLS 2024)* (pp. 131–138). International Society of the Learning Sciences.
8. Xia, F., Schenck, K. E., Swart, M. I., Grondin, M., Kim, D., Kwon, O. H., Walkington, C., & Nathan, M. J. (2024). How pedagogical hints impact embodied geometric reasoning. *Proceedings of the 18th International Conference of the Learning Sciences (ICLS 2024)*, 123–130.
9. Kim, D., & Nathan, M. J. (2023). Exploring the link between cognitively relevant actions and geometric reasoning: A pilot study. In P. Blikstein, J. Van Aalst, R. Kizito, & K. Brennan (Eds.), *Proceedings of the 17th international conference of the learning sciences (ICLS 2023)* (pp. 834–837). International Society of the Learning Sciences.
10. Sung, H., Kim, D., Swart, M. I., & Nathan, M. J. (2023). Multimodal behavior analysis: Two patterns of collaborative construction of embodied knowledge. In M. Goldwater, F. K. Anggoro, B. K. Hayes, & D. C. Ong (Eds.), *Proceedings of the 45th annual meeting of the cognitive science society* (Vol. 45, pp. 1587–1592).
11. Swart, M. I., Schenck, K. E., Xia, F., Kim, D., Grondin, M. M., & Nathan, M. J. (2023). Embodying students' geometric thinking in an interactive narrative game. *Proceedings of the 2023 AERA Annual Meeting*.
12. Grondin, M. M., Kim, D., Raymond, C., Pendey, A., De Leon, C., Morrell, M., Swart, M. I., Xia, F., & Nathan, M. J. (2023). Embodying engineering students' knowledge through gesture. *Proceedings of the 2023 AERA Annual Meeting*.
13. Schenck, K. E., Kim, D., Swart, M. I., & Nathan, M. J. (2022). With no universal consensus, spatial system perspective affects model fitting and interpretation for mathematics. *Proceedings of the 2022 AERA Annual Meeting*.
14. Xia, F., Swart, M. I., Kim, D., & Nathan, M. J. (2021). Eliciting predictive behaviors to support embodied mathematical cognition: Socially distanced experimentation in the time of COVID-19. *Proceedings of the 2021 AERA Annual Meeting*.
15. Swart, M. I., Schenck, K. E., Xia, F., Kim, D., Kwon, O. H., Nathan, M. J., & Walkington, C. (2021). Embodiment as a Rosetta Stone: Collective conjecturing in a multilingual classroom using a motion capture geometry game. In A. I. Sacristán, J. C. Cortés-Zavala, & P. M. Ruiz-Arias (Eds.), *Mathematics education across cultures: Proceedings of the 42nd meeting of the north american chapter of the international group for the psychology of mathematics education* (pp. 2243–2251). PMENA.
16. Park, W., Kim, D., & Kang, D. Y. (2021). Research trends in science and mathematics education in South Korea 2014–2018: A cross-disciplinary analysis of publications in selected local journals. *Asia-Pacific Science Education*, *7*(2), 280–308.
17. Kirankumar, V., Sung, H., Swart, M., Kim, D., Xia, F., Kwon, O. H., & Nathan, M. (2021). Embodied transmission of ideas: Collaborative construction of geometry content and mathematical thinking. In C. E. Hmelo-Silver, B. De Wever, & J. Oshima (Eds.), *Proceedings of the 14th international conference on computer-supported collaborative learning 2021* (pp. 177–180). International Society of the Learning Sciences.
18. Kim, D., Swart, M. I., Schenck, K. E., & Nathan, M. J. (2021). Grounded and embodied proof production: Are gestures and speech enough to produce deductive proof? In E. de Vries, Y. Hod, & J. Ahn (Eds.), *Proceedings of the 15th international conference of the learning sciences (ICLS 2021)* (pp. 1109–1110). International Society of the Learning Sciences.
19. Kim, D., & Kwon, O. N. (2019). Moving fingers and the density of rational numbers: An inclusive materialist approach to digital technology in the classroom. In P. Drijvers, K. M. Clark, & S. Rezat (Eds.), *Proceedings of the 11th congress of the european society for research in mathematics education* (pp. 2868–2875). Utrecht University; ERME.
20. Kim, D., & Kwon, O. N. (2018). How dense are rational numbers?: A new materialist approach to the number line. In E. Berqvist, M. Österholm, C. Granberg, & L. Sumpter (Eds.), *Proceedings of the 42nd conference of the international group for the psychology of mathematics education* (Vol. 5, pp. 85–85).
21. Kim, D., & Kwon, O. N. (2018). How dense are rational numbers?: An inclusive materialist case study to digital technology. *Education of Primary School Mathematics*, *21*(4), 375–395.
22. Han, C., Lee, K., Kim, D., Bae, M. S., & Kwon, O. N. (2018). Aspects of understandings on statistical variability across varying degrees of task structuring. *Education of Primary School Mathematics*, *21*(2), 131–150.
23. Han, C., Bae, M. S., Kim, D., Lee, K., & Kwon, O. N. (2017). Degree of task structuring for developing the notion of variability. In B. Kaur, W. K. Ho, T. L. Toh, & B. H. Choy (Eds.), *Proceedings of the 41st conference of the international group for the psychology of mathematics education* (Vol. 1, pp. 205–205).

24. Yoon, J. E., Kim, D., & Kwon, O. N. (2015). Teachers' roles of learner-centered classes in domestic mathematics education research. *Journal of Learner-Centered Curriculum and Instruction*, *15*(1), 45–68.

Other

1. Kim, D., Beliakoff, A., & Strother, S. (2025). Impact evaluation of a virtual reality program on students' geometry achievement. *SREE 2025 Conference: Education in Context – Research, Systems, and the Future of Evidence-Based Change*.

Awards, Fellowships & Honors

Jun 2026	International Society of Learning Sciences Doctoral Consortium	*International Society of the Learning Sciences*
Jun 2026	Immersive Learning Research Network Doctoral Colloquium	*Immersive Learning Research Network*
Sep 2025–Present	Arvil S. Barr Fellow	*School of Education – University of Wisconsin-Madison*
Sep 2021–Present	Interdisciplinary Training Program International Affiliate	*University of Wisconsin-Madison*
Oct 2021, 2023, 2024	Educational Psychology Travel Award (×3)	*Department of Educational Psychology – University of Wisconsin-Madison*
Aug 2019–May 2023	Graduate School Fellow	*Graduate School – University of Wisconsin-Madison*
Sep 2019–May 2023	Kwanjeong Fellow	*Kwanjeong Educational Foundation*
Mar 2009–Feb 2011	National Science and Engineering Undergraduate Scholarship	*Korean Student Aid Foundation*

Skills

Advanced prompt engineering · AI-assisted data visualization & statistical programming	*Generative AI*
Interview & observation protocols · Discourse & video analysis · Epistemic Network Analysis (ENA) · Qualitative coding (MAXQDA)	*Qualitative Research*
Experimental & quasi-experimental design · Generalized linear modeling (logistic; Poisson) · Hierarchical linear modeling · Causal inference · Structural equation modeling · Item response theory · Principal component & factor analysis · Meta-analysis & integrative data analysis · Data wrangling & exploratory analysis · Data visualization	*Quantitative Research*
Research team supervision & mentorship · Undergraduate & junior graduate mentorship · Project management	*Research Leadership*
R (statistical programming & Quarto) · Qualtrics · MAXQDA · ENA Web Tool	*Research Software*

Leadership, Mentoring & Service

Research Mentorship

iGRASP Research Team

Team Lead *Jan 2026–Present*

- Developed structured onboarding and training materials
- supervised 1 junior graduate student and 6 undergraduate research assistants in experiment administration and scientific data collection
- maintained team culture through weekly group meetings and biannual individual check-ins
- supported team members in translating research experience into professional assets (resume and LinkedIn)

iGRASP Research Team

FOUNDER & TEAM LEAD *Sep 2025–Dec 2025*

- Founded, named, and branded the iGRASP research team
- created team identity and produced training materials
- onboarded 1 junior graduate student and 2 undergraduate research assistants
- focus on research design, instrument development, and implementation

GRAID Research Internship – UW-Madison

RESEARCH MENTOR *Jan 2025–May 2025*

- Mentored an undergraduate research intern on data management and quantitative data analysis in social sciences research embedded within the GRAID project

SERP Research Internship – UW-Madison

RESEARCH MENTOR *Jun 2020–Aug 2020*

- Mentored a first-generation undergraduate student through the Summer Education Research Program (SERP)
- introduced foundational skills in reading and critically evaluating academic literature and core concepts in educational research

PEER REVIEW

Ongoing	European Journal of Psychology of Education
Ongoing	Computers and Education: X Reality (Special Issue)
Ongoing	American Educational Research Association Annual Meetings
Ongoing	International Society of the Learning Sciences Annual Meetings
Ongoing	North American Chapter of PME Annual Meetings

COMMITTEE SERVICE

Korean EdTech and Learning Sciences Research Network

BOARD MEMBER *Mar 2026–Present*

- Curated academic resources on educational technology and learning sciences development and research for members

Educational Psychology Student Association – UW-Madison

CO-CHAIR OF ACADEMIC EVENTS *Sep 2021–May 2022*

- Organized academic events and professional development opportunities for graduate students in the department, including EPSA Brown Bag sessions, invited talks, and informal faculty chats

PRESENTATIONS & WORKSHOPS

Apr 2025	Speaker	*How I Did My Dissertation Proposal – UW-Madison*
Feb 2025	Speaker	*How I Did My Preliminary Examination – UW-Madison*
May 2024	Instructor	*How to Do Data Science with Quarto Markdown – MAGIC Lab*
Mar 2024	Instructor	*How to Do Qualitative Data Analysis with MAXQDA2024 – MAGIC Lab*
Feb 2024	Instructor	*How to Organize Literature Notes with Obsidian – UW-Madison*
Sep 2023	Instructor	*Introduction to Zotero – UW-Madison*
Oct 2023	Speaker	*Writing as a Non-Native English Speaker – UW-Madison*
Sep 2023	Speaker	*Internship for Graduate Students – UW-Madison*
Mar 2021	Instructor	*How to Analyze Speech and Video Data Using Transana – MAGIC Lab*
2021–2025	Panelist	*UW Department of Educational Psychology Annual Campus Visit Day*

Panel Discussions

2021–2025	Panelist	*UW Department of Educational Psychology Annual Campus Visit Day*
2021–2024	Panelist	*EPSA Brown Bag – Senior Graduate Student Panel*
Mar 2024	Panelist	*Demystifying the Preliminary Exams – UW-Madison*